

What are the factual environmental impacts of AI usage?

Posted by u/Secure-Evening • 0 points • 10 comments

I feel like this is never brought up to be discussed in any serious way. I'm looking to hear from both anti and pro-AI people here. Please cite your sources if you're making a factual statement. I hear things like "It takes 2L of water per prompt" and "It's only the equivalent of 10 google searches" though never any sources behind them.

Comments

u/oneonus • 1 points • 2025-09-21 01:49 UTC

AI data centres are ruining the environment, taking all of our water and are power hungry.

Case in point:

<https://techiegamers.com/texas-data-centers-quietly-draining-water/>

Larger data centers can each "drink" up to 5 million gallons per day, or about 1.8 billion annually, usage equivalent to a town of 10,000 to 50,000 people.

<https://www.eesi.org/articles/view/data-centers-and-water-consumption>

In addition, data centers are power hungry and result in sound drone for surrounding areas, as well causing local utilities rate to spike by 50%. Following is outstanding work by Business Insider.

[Exposing the Dark Side of AI Data Centers](<https://youtu.be/t-8TDOFqkQA?feature=shared>)

Business Insider, in addition, also discovered that the power needs of data centers have forced some states to withdraw from their carbon emissions targets. Power companies are even looking to extend the life of coal and gas plants to help meet the unprecedented demand. Not to mention massive tax breaks for the richest companies.

u/4215-5h00732 • 1 points • 2025-06-30 17:41 UTC

The information I've seen about emissions comes from a 2019 paper.

<https://arxiv.org/abs/1906.02243>

MIT review:

https://www.technologyreview.com/2019/06/06/239031/training-a-single-ai-model-can-emit-as-much-carbon-as-five-cars-in-their-lifetimes/#amp_tf=From%20%251%24s&aoh=17513050144485&referrer=https%3A%2F%2Fwww.google.com&share=https%3A%2F%2Fwww.technologyreview.com/2019/06/06/239031/training-a-single-ai-model-can-emit-as-much-carbon-as-five-cars-in-their-lifetimes%2F

u/Beautiful-Lack-2573 • 7 points • 2025-06-29 21:44 UTC

These are all sourced and offer good starting points:

<https://andymasley.substack.com/p/a-cheat-sheet-for-conversations-about>

<https://www.iea.org/reports/energy-and-ai>

<https://journal.uptimeinstitute.com/water-is-local-generalities-do-not-apply/>

But since you asked about the environment:

Power consumption isn't a bad thing, but emissions are. All the misleading stats are about electricity, not about emissions. Electricity is not scarce and electricity use is only going to go up. We are building infrastructure to generate massively more electricity across the world, mostly for decarbonization and BEVs.

Similarly, water usage isn't a bad thing, but water shortages are. Globally, water is abundant. Locally, it's often not.

Data centers are eyesores and built in low-income areas and have negative externalities, just like power plants or waste recycling facilities.

u/4215-5h00732 • 1 points • 2025-06-30 17:32 UTC

Power consumption can be a bad thing if the grid can not handle it. CA SMUD has programs that give them the ability to pause your HVAC usage if needed. They do this specifically to help maintain the stability of the grid.

Electricity is scarce in the US at least. It's the classic "supply and demand" problem.

Why electricity prices are surging for U.S. households <https://share.google/YsOqrT5IU9w4S7RFj>

Sticking with the US, ~31% of the lower 48 are in a drought. It's great if water is plentiful across the globe, but I'm not aware how that helps. I'm also not convinced that's even true. Do you have a source?

Water Facts - Worldwide Water Supply | ARWEC| CCAO | Area Offices | California-Great Basin | Bureau of Reclamation
<https://share.google/rkSwj4DCGa5XTLeOT>

u/Eastern-Zucchini6291 • 2 points • 2025-06-30 07:46 UTC

The biggest misinformation I've seen is lumping in data centers and AI together. And using comparison, like phone charges (which take like nothing to charge).

u/Afraid-Ad8702 • -4 points • 2025-06-29 21:31 UTC

Unpopular opinion, call me a conspirationist but I personally couldn't care less, imo the whole carbon footprint topic is a government scam to make us pay more taxes and strip us of our freedom and most basic rights.

u/Secure-Evening • 5 points • 2025-06-29 21:53 UTC

That's just factually incorrect and you have to have your head in the sand to believe that. Especially since they barely care about environmentalism anyways. We have to fight to get them to pass legislation that helps support the environment or to stop spending a bunch of tax payer dollars on fossil fuel companies. So that makes no sense.

u/Gimli • 2 points • 2025-06-29 20:47 UTC

Wildly different depending on who runs it, where, on what hardware, using what models...

AI is just computing. Computation uses electricity. That's it.

Depending on where you are in the world, power may be fossils, renewables, or a mix.

Depending on where you are in the world, cooling might be free (cold places), or hard (hot places). Water may be abundant or extremely scarce. Water may be needed or not needed.

In general though, LLMs are much hungrier than image generators.

u/Far-Fennel-3032 • 3 points • 2025-06-29 20:45 UTC

To put it simply image generation models can be run locally on your computer, and the whole big deal of deepseek was a LLM could be put on a powerful local computer.

Generally speaking people don't understand the scale of energy and water use of energy day life. The following article on nature has a great breakdown on the assorted comparisons of a person doing something and a LLM. It was published on Nature which has extremely high standards to get something published there.

<https://www.nature.com/articles/s41598-024-76682-6>

u/KamikazeArchon • -2 points • 2025-06-29 20:30 UTC

This is not a good place to get such information. Reddit has no mechanism to evaluate the accuracy or reputability of a source.

There are scientific journals, industry watch groups, etc. that gather information and subject it to statistical analysis, peer review, and such things. You would be better served looking to them for answers.

u/DeepFollowing9403 • 2 points • 2025-06-29 22:08 UTC

Right... and you can find that information on reddit.

u/Redz0ne • 0 points • 2025-06-29 20:41 UTC

Because reddit totally isn't a hugbox.